



AMERICAN INTERNATIONAL UNIVERSITY BANGLADESH

LAB COVER SHEET

Experiment Title:	Implementation of Asynchronous and synchronous counter basic using flip-flops.		
Experiment No:	07	Date of Submission:	24-09-2024
Course Title:	DIGITAL LOGIC AND CIRCUITS LAB		
Course Code:	COE3103	Section: L	
Semester:	Summer 2023-24	Course Teacher: DR. MD. KABIRUZZAMAN	

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FACULTY COMMENTS	Marks Obtained	
	Total Marks	

Title: Implementation of Asynchronous and synchronous counters using flip-flops.

Introduction:

Counters are combinations of flip-flops arranged so that they can remember how many clock pulses have been applied over some specified interval. The flip-flops are often interconnected so that only a portion of their available binary states can be supported. If there are N flip-flops being used in a counter, the number of states available is $2N$. If the counter proceeds cyclically through K of these states, where $K \leq 2N$, it is said to be a modulo K (or MOD K) counter. Some applications will require a separate output to indicate each of the counters states, alternatively, other applications may require only one output pulse every K th state. Counters are classified into two broad categories according to the way they are clocked:

asynchronous and synchronous. In asynchronous counters, commonly called ripple counters, the first flip-flop is clocked by the external clock pulse and then each successive flip flop is clocked by the output of the preceding flip-flop. In synchronous counters, the clock input is connected to all of the flip-flops so that they are clocked simultaneously. The objective of this experiment is designing of the following counters using J-K Flip-Flops (IC 74LS76)

(a) n -bit Binary Asynchronous Counter

(b) n -bit Binary Synchronous Counter

Theory and Methodology:

Asynchronous counter:

A three-bit asynchronous counter is shown in the figure 9.1. The external clock is connected to the clock input of the first flip-flop (FF0) only. So, FF0 changes state at the falling edge of each clock pulse, but FF1 changes only when triggered by the falling edge of the Q output of FF0. Because of the inherent propagation delay through a flip-flop, the transition of the input clock pulse and a transition of the Q output of FF0 can never occur at exactly the same time. Therefore, the flip-flops cannot be triggered simultaneously, producing an asynchronous operation. Figure 9.1 gives a three-bit counter capable of counting from 0 to 7. The clock inputs of the three flip-flops are connected in cascade. The T input of each flip-flop is connected to a constant 1, which means that the state of the flip-flop will be reversed (toggled) at each positive edge of its clock. We are assuming that the purpose of this circuit is to count the number of pulses that occur on the primary input called Clock. Thus, the clock input of the first flip-flop is connected to the Clock line. The other two flip-flops have their clock inputs driven by the Q output of the preceding flip-flop. Therefore, they toggle their state whenever the preceding flip-flop changes its state from $Q = 1$ to $Q = 0$, which results in a positive edge of the Q signal. Figure 9.1 shows a timing diagram for the counter. The value of Q_0 toggles once each clock cycle. The change takes place shortly after the positive edge of the Clock signal. The delay is caused by the propagation delay through the flip-flop. Since the second flip-flop is clocked by Q_0 , the value of Q_1 changes shortly after the negative edge of the Q_0 signal. Similarly, the value of Q_2 changes shortly after the negative edge of the Q_1 signal. If we look at the values $Q_2Q_1Q_0$ as the count, then the timing diagram indicates that the counting sequence is 0, 1, 2, 3,

4, 5, 6, 7, 0, 1, and so on. This circuit is a modulo-8 counter. Because it counts in the upward direction, we call it an up-counter.

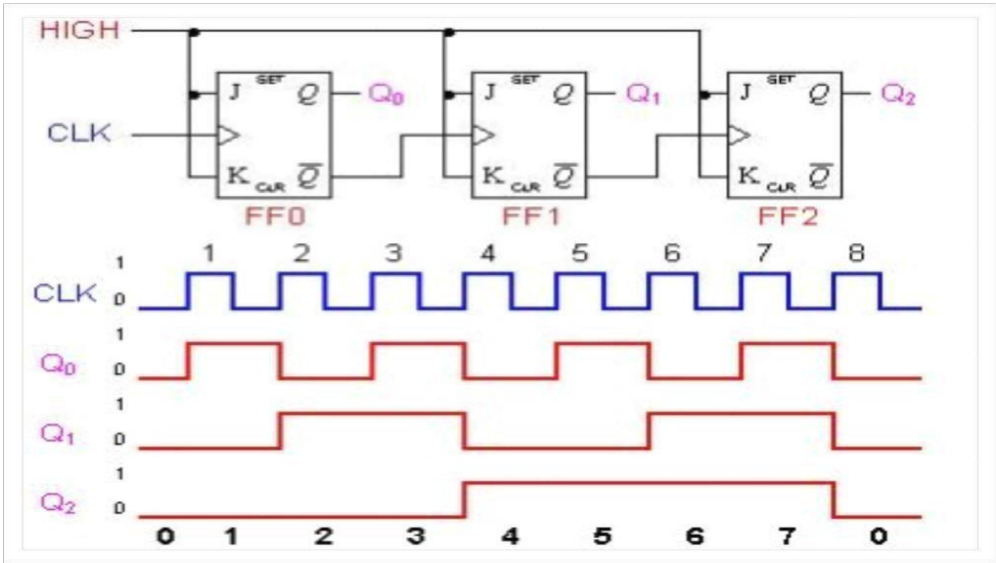


Figure 9.1: 3 bit Asynchronous counter and its timing diagram

Synchronous counter

In synchronous counters, the clock inputs of all the flip-flops are connected together and are triggered by the input pulses. Thus, all the flip-flops change state simultaneously (in parallel). The most important advantage of synchronous counters is that there is no cumulative time delay because all flip-flops are triggered in parallel. Thus, the maximum operating frequency for this counter will be significantly higher than for the corresponding ripple counter.

Table 9.1 shows the contents of a four-bit up-counter for eight consecutive clock cycles, assuming that the count is initially 0. Observing the pattern of bits in each row of the table, it is apparent that bit Q0 changes on each clock cycle. Bit Q1 changes only when Q0 = 1. Bit Q2 changes only when both Q1 and Q0 are equal to 1. In general, for an n-bit up-counter, a given flip-flop changes its state only when all the preceding flip-flops are in the state Q = 1.

Table 9.1

Clock cycle	Q ₂	Q ₁	Q ₀	
0	0	0	0	
1	0	0	1	
2	0	1	0	Q ₁ changes
3	0	1	1	
4	1	0	0	Q ₂ changes
5	1	0	1	
6	1	1	0	
7	1	1	1	
8	0	0	0	



0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

There are 2 JK Flip Flops in one IC. Here

1. *Journal of the American Medical Association*, 2000; 284: 2689-2695.



IC 7408 contains 4 AND gates in it. The pin configuration is shown below:

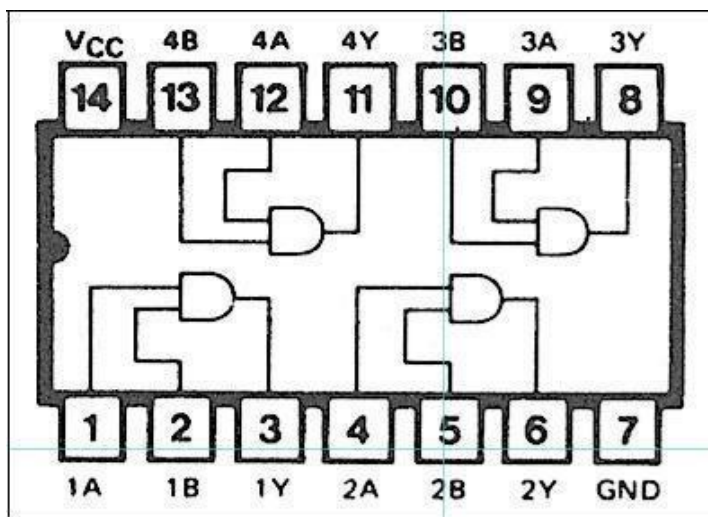


Figure 9.3: IC 74LS76

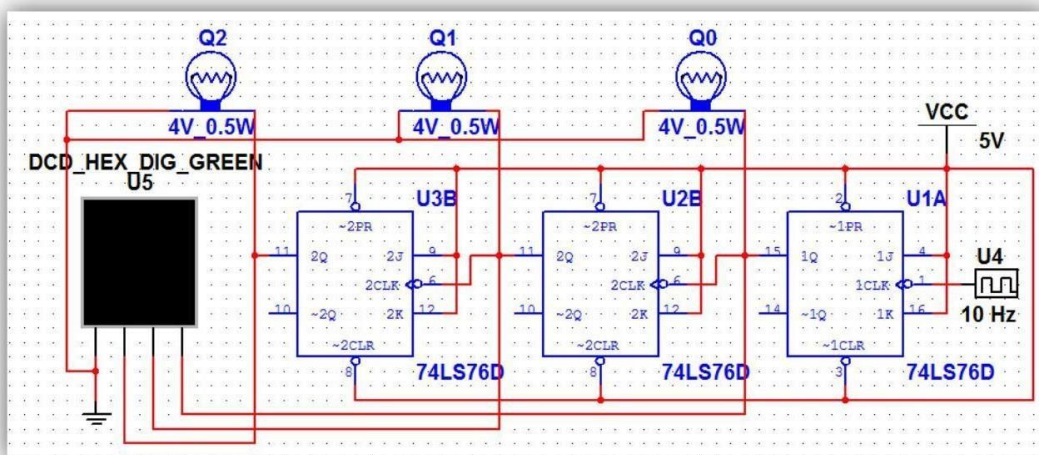


Fig 9.5: 3 bit Asynchronous Counter

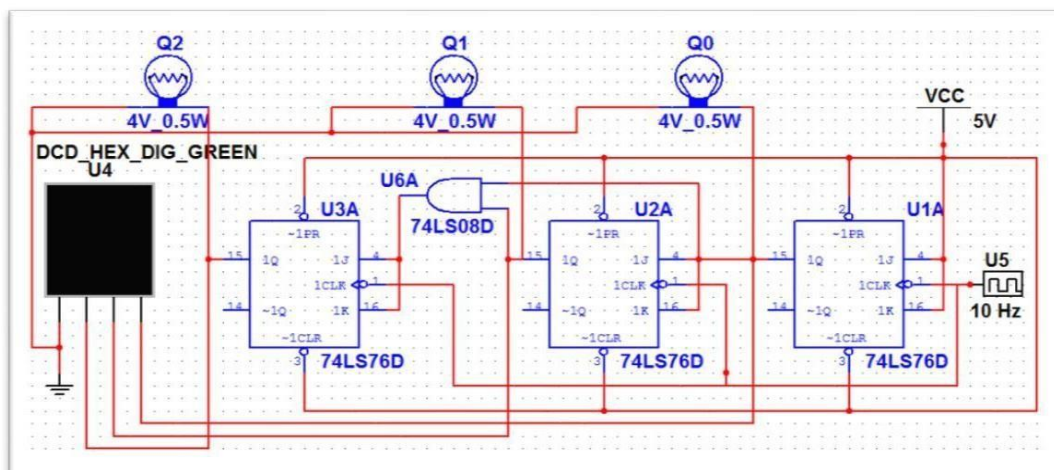


Fig 9.6: 3 bit Synchronous Counter

Apparatus:

- IC 74LS76 (JK Flip Flop)
- IC 7408 (AND Gate)
- LED Lamps or Display
- Trainer Board
- Oscilloscope
- Connecting Wires

Precautions:

- Before preparing the circuits, check all the ICs(74LS76 & 7408) to make sure they are all working properly.
- Careful about the biasing of JK Flip Flops.
- Make sure that connected the preset and clear pins with Vcc.
- Try to use as less wire as possible. Make sure there is no loose connection.
- If use LED display, then make sure the proper biased.
- For Clock pulse, the trainer board's analog signal generator is a good choice. Use lower frequencies so that the change is slow enough to observe the outputs and take readings.

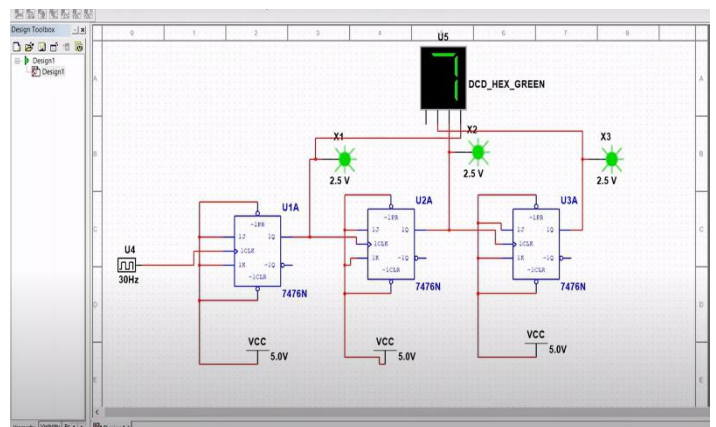
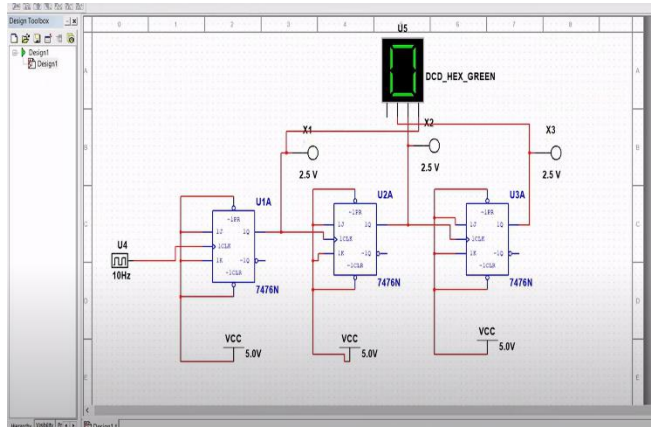
Experimental Procedure:**Part 1: 3 bit Asynchronous Counter**

1. The circuit was designed on the bread board as shown in Figure 9.5.
2. The trainer board's signal generator was used for the clock pulse and power supply for biasing the Flip Flops.
3. The output was also viewed in oscilloscope; outputs were connected to the different channels of the oscilloscope.
4. The output results were observed, results were recorded and also took some pictures for our lab report.
- 5.

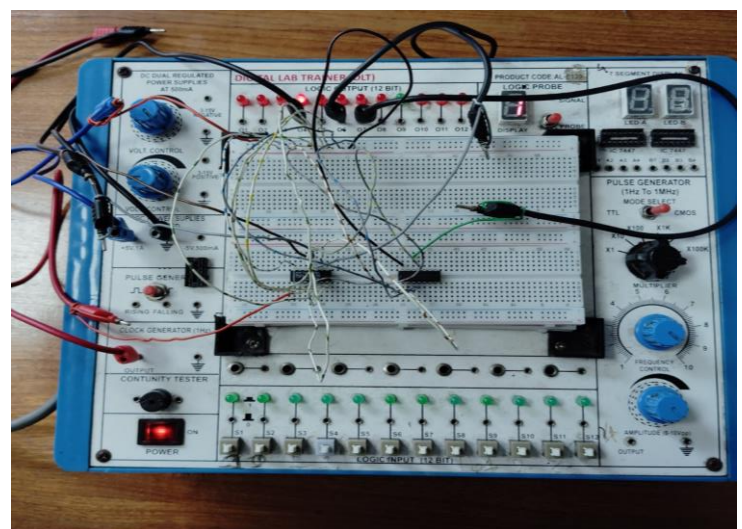
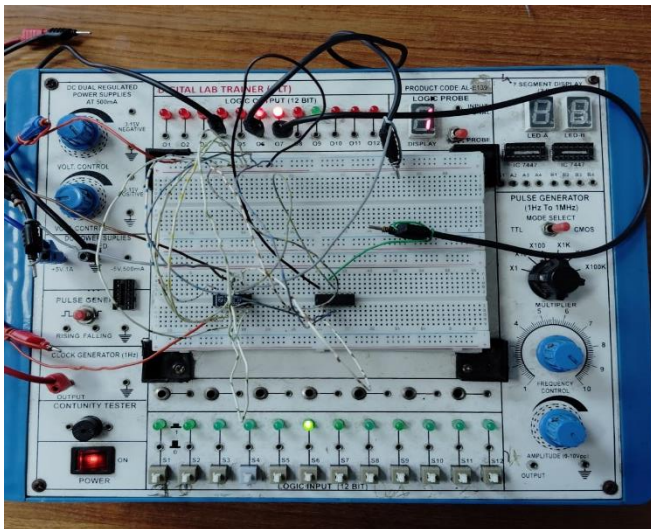
3 bit Synchronous Counter

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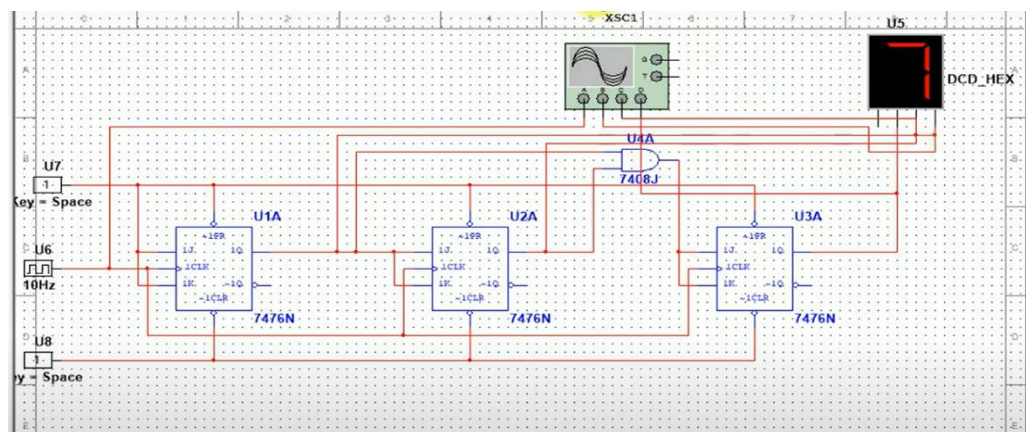
Simulation and Measurement:



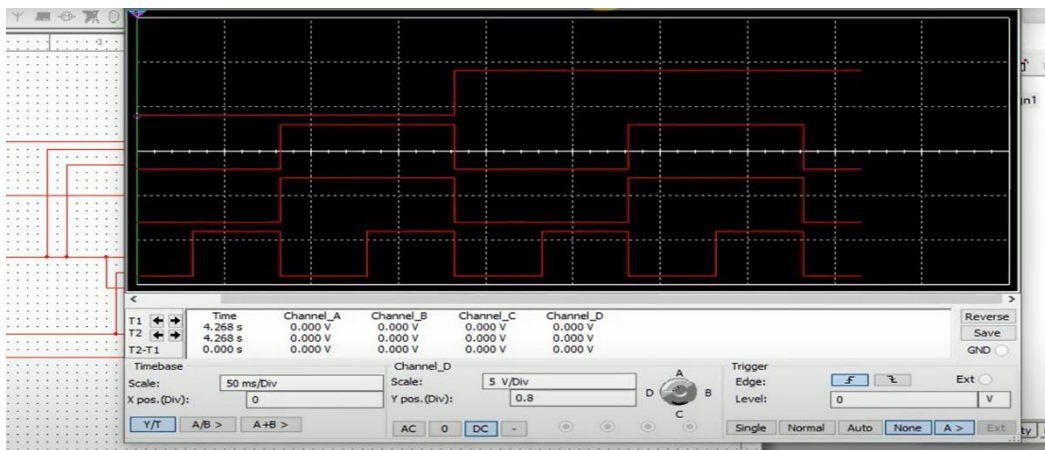
3 bit Asynchronous Counter



Lab Image



3 bit Synchronous Counter



3 bit Synchronous Counter

Truth Table:

CLOCK PULSE	Q2	Q1	Q0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8 recycles	0	0	0

The simulation results with the experimental data/ wave shapes are same. There are no difference between experimental data and simulation results.

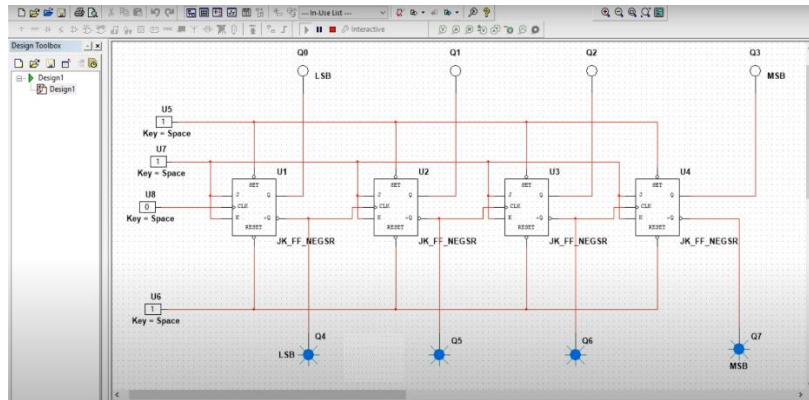
Discussion:

In asynchronous counters, each flip-flop's clock input is connected to the output of the preceding flip-flop. This creates a ripple effect, where the state of the counter changes one flip-flop at a time. Simple design and implementation. Propagation delays between flip-flops can lead to glitches and inaccurate counting. This becomes more pronounced with larger counters. In synchronous counters, all flip-flops are clocked simultaneously by a common clock signal. The state of each flip-flop is determined by a combinatorial logic circuit that takes the current state and the input signal as inputs. Faster and more accurate than asynchronous counters, especially for larger counts. No propagation delays between flip-flops. More complex design and implementation, requiring additional logic for state control.

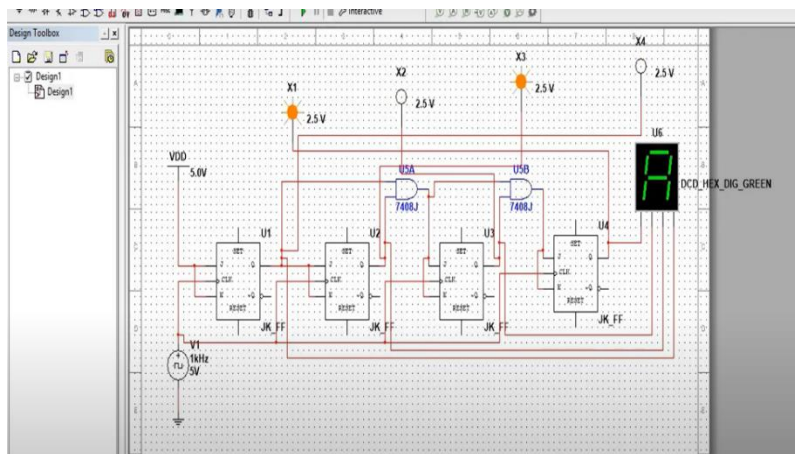
Conclusion:

Asynchronous counters are suitable for applications where speed and accuracy are not critical, and simplicity is a priority. Synchronous counters are preferred in applications where high-speed and accurate counting are essential, such as digital clocks, frequency counters, and data transmission systems.

Questions with answers for report writing:

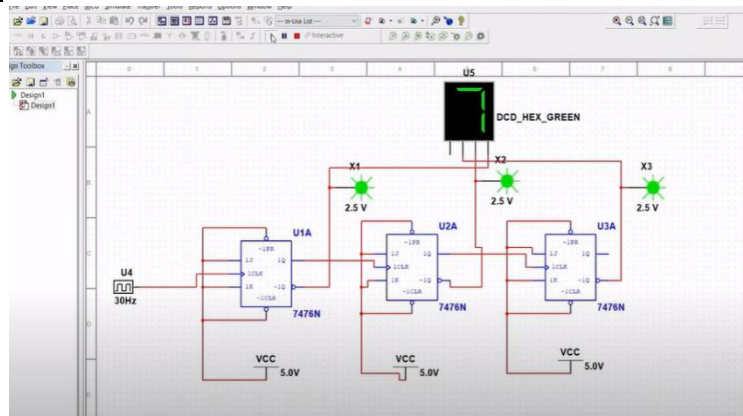


4 bit Asynchronous Up- Counter

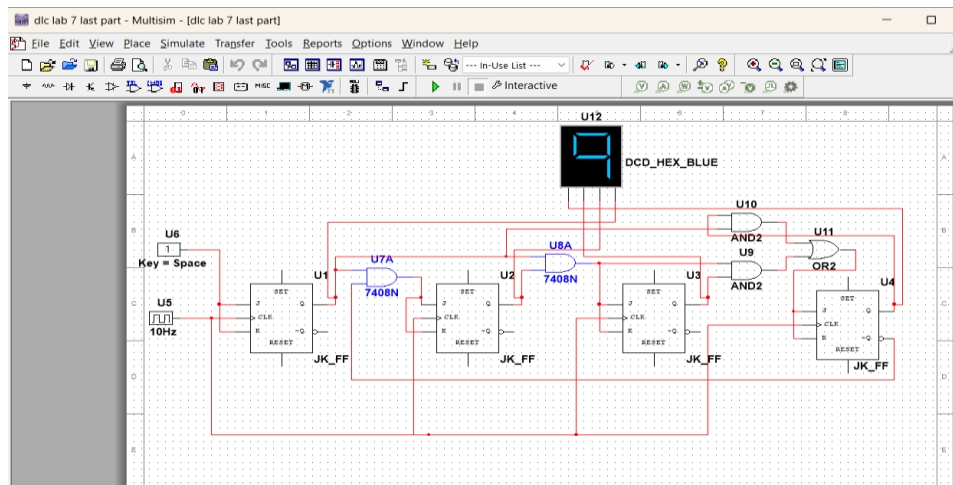


4 bit Synchronous Up- Counter

Bonus Mark:



3 bit Asynchronous down counter



Mod 10 Synchronous up counter

Reference(s):

- i) Thomas L. Floyd, "Digital Fundamentals", Ninth Edition.
- ii) <https://www.geeksforgeeks.org/differences-between-synchronous-and-asynchronous-counter/>
- iii) <https://www.shiksha.com/online-courses/articles/difference-between-synchronous-and-asynchronous-counter/>
- iv) <https://www.geeksforgeeks.org/design-mod-n-synchronous-counter/>